

LaTeX Setup Guide

There are two common ways to set up a LaTeX environment:

1. **Overleaf** — Online LaTeX environment
2. **VS Code** — Local LaTeX environment

1. Overleaf

Configuration

Overleaf is a browser-based LaTeX editor. No local LaTeX installation is required.

Step 1: Create an Overleaf Account

Go to [SJTU_Overleaf](#) and use JAccount to sign in.

Step 2: Create a Project

After logging in:

New Project → Blank Project

Give the project a name and create it.

Step 3: Edit the `.tex` File

Overleaf automatically creates a main `.tex` file. For example:

```
\documentclass{article}
```

```
\title{LaTeX wksp}
```

```
\author{Your name}
```

```
\date{\today}
```

```
\begin{document}
```

```
\maketitle
```

```
\section{Introduction}
```

```
\end{document}
```

Step 4: Compile

Click **Recompile** to compile the document and generate the PDF.

The PDF preview will appear directly in the Overleaf interface.

Note: For Chinese character, use XeLaTeX compiler and add `\usepackage{ctex}`

How to Select the Compiler in Overleaf:

1. Open your Overleaf project.
2. Click Menu in the top-left corner.
3. Find Compiler.
4. Change the compiler from:
pdfLaTeX to XeLaTeX
5. Close the menu and click Recompile.

```
\documentclass{article}
\usepackage{ctex}

\title{中文 title}
\author{Your name }
\date{\today}

\begin{document}

\maketitle

\section{Introduction}

\end{document}
```

2. VS Code

Configuration

A local LaTeX environment requires:

```
VS Code
+
LaTeX Workshop (Extension in VSCode)
+
LaTeX Distribution (Download from Official website)
```

For Windows, **MiKTeX** or **TeX Live** can be used as the LaTeX distribution.

Step 1: Install VS Code

Download and install [Visual Studio Code](#).

Step 2: Install a LaTeX Distribution

Choose one of the following:

- [MiKTeX](#) — relatively lightweight and beginner-friendly
- [TeX Live](#) — more comprehensive and includes a large collection of LaTeX packages

Install the selected distribution using its default settings unless you have specific requirements.

Step 3: Install LaTeX Workshop

Open VS Code and go to:

Extensions → Search "LaTeX Workshop"

Install **LaTeX Workshop** by James Yu.

Step 4: Create a .tex File

Create a file named:

main.tex

For example:

```
\documentclass{article}
```

```
\begin{document}
```

```
Hello, LaTeX!
```

```
\section{Introduction}
```

```
This is my first LaTeX document.
```

```
\end{document}
```

Step 5: Compile

Open `main.tex` in VS Code and use the **Build LaTeX project** command provided by LaTeX Workshop.

The generated PDF can be viewed directly in VS Code.

3. Comparison

Feature	Overleaf	VS Code + Local LaTeX
Setup difficulty	★ Very easy	★ ★ ★ More complicated
Offline work	✗	✓
Customization	Limited	Highly customizable
Package management	Managed by Overleaf	User-managed
Cross-device access	★ ★ ★ ★ ★ Excellent	★ ★ ★ Requires environment setup
Multiple user	✗	✓